1. Overview of Valuation

MGMT 648: Valuation

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Today's Session

Why Valuation Matters

- Project vs. enterprise valuation
- Value creation and NPV
- Why big investments fail

Five Key Issues

- Does the story make sense?
- Risks, financing, earnings
- Flexibility and follow-on options

Process, Bias, and AI Tools

- Three-phase evaluation process
- Decision bias and checks
- Building valuation models with Claude

Why Valuation Matters

The Fundamental Question

Project Valuation

A firm assembles assets to create productive capacity.

- New plant, equipment, product line
- Technology investment
- Entering a new market

The question: is the value of the future cash flows worth more than the cost of building?

Enterprise Valuation

A firm acquires an existing business.

- Mergers and acquisitions
- Buyouts, divestitures
- Strategic partnerships

The question: is the target worth more than what we pay — including any synergies?

The tools and principles are the same in both cases. This course covers both.

Value Creation

A firm has the opportunity to invest $100 million in a project that generates cash flows valued at $150 million.

- Value created = $150M − $100M = **$50 million**
- In finance jargon: the project has a **net present value (NPV) of $50 million**
- The investment makes the firm's shareholders $50 million wealthier

> The entire valuation enterprise is aimed at answering one question accurately: **Is the present value of expected future cash flows greater than the cost of the investment?**

Why Do Big Investments Fail?

The Record

- Over half of all large investment projects fail to achieve hoped-for results
- Daimler-Benz acquired Chrysler for $35B in 1998 — sold for $7.4B in 2007, then bankruptcy
- M&A value destruction is the rule, not the exception

The Causes

- Risky projects sometimes just fail — that is expected
- But many failures stem from **flawed analysis**, not just bad luck
- Complexity and uncertainty push managers toward gut instinct
- Analytical tools + disciplined process improve the odds

Good intuition matters. But intuition plus rigorous analysis beats intuition alone, especially on large irreversible commitments.

Five Key Issues

Every major investment decision involves these five questions

The Five Issues

#	Issue	The Core Question
1	Does the story make sense?	Does the firm have a real competitive advantage here?
2	What are the risks?	What could go wrong, and how bad could it get?
3	How can it be financed?	What financing is available, and does it add or destroy value?
4	Effect on near-term earnings?	Will this dilute EPS in the short run, and does that matter?
5	Inherent flexibilities?	Can it be staged? Are there valuable follow-on options?

These issues frame the qualitative assessment that must accompany — and often drive — the financial model.

Issue 1: Does the Story Make Sense?

Competitive Advantage

Does the firm have an edge — specialized knowledge, technology, relationships, or scale — that others cannot easily replicate?

- If the answer is no, competition will erode returns
- A positive NPV that any competitor can also earn is not a durable positive NPV

Comparative Advantage

Is *this firm* better positioned to capture the opportunity than rivals?

- What capabilities does the firm bring that others lack?
- Are those advantages sufficient to deter competitors from making similar investments?

The story must be specific and honest. Vague claims like "we are well-positioned in a growing market" do not constitute a competitive advantage

Issue 2: What Are the Risks?

A careful assessment of **what might go wrong** is often more valuable than a projection of what might go right.

Risk Identification

- Market and demand risk
- Technology and execution risk
- Regulatory and political risk
- Counterparty and financing risk
- Competitive response

Risk Management

- How should risks be incorporated into the analysis?
- Which risks affect the discount rate vs. the cash flow forecasts?
- Are there insurance, hedging, or contractual tools that can transfer risk?

Financing is not just a funding question — it can be a **source of value** in its own right.

Internal vs. External

- Can the project be funded from operating cash flows?
- Or does it require external debt or equity?
- Financing constraints affect which projects get pursued

Debt Structure

- On-balance-sheet debt vs. project finance (nonrecourse)
- Government guarantees or credit enhancements
- Special-purpose entities for off-balance-sheet treatment

Value Impact

- Favorable financing terms can materially raise NPV
- The ability to transfer project risk through financing is itself a source of value
- Financing constraints can kill otherwise good projects

Issue 4: Effect on Near-Term Earnings?

Dilutive Investments

Large upfront expenses + deferred cash flows = depressed EPS in early years

- Common in oil, pharma, infrastructure
- Can make managers reluctant to invest even when NPV is positive
- Especially acute when bonuses are tied to EPS

Accretive Investments

Investment adds to EPS immediately or quickly

- Acquisitions that add profitable revenue
- Cost-saving projects that reduce expenses in Year 1
- Managers may over-favor accretive deals even at high prices

Neither dilutive nor accretive is inherently good or bad. What matters is NPV — but earnings effects shape managerial incentives and can distort investment decisions

Issue 5: Inherent Flexibilities?

Staged Investment

Can the investment be broken into phases, with each stage contingent on the results of the prior one?

Staging acquires the **option to proceed** — or walk away — as information arrives.

Follow-On Opportunities

Does this investment open doors to future opportunities?

Early-stage investments in new markets or technologies often derive most of their value from the follow-on options they create — not the immediate cash flows.

Production & Marketing Synergies

Does the new investment share production or marketing assets with existing operations?

Shared infrastructure or distribution channels can give an acquirer or investor a real cost or revenue advantage.

Applying the Framework: Cisco Acquires Linksys (2003)

Purchase price: ~$500 million

Issue	Cisco's Assessment
Does the story make sense?	Consumer broadband adoption making home networking a mass market; Linksys is the recognized leader. Extends Cisco's end-to-end portfolio into high-growth segments.
What are the risks?	Wireless technology risk — could a new standard render Linksys products obsolete? Mitigated by acquiring the industry leader.
How can it be financed?	Cisco issues common stock — avoids cash drain, aligns seller incentives with combined firm.
Effect on near-term earnings?	Accretive to GAAP and pro forma EPS by $0.01 in FY2004.
Inherent flexibilities?	Platform acquisition — entry into SOHO networking opens substantial follow-on investment opportunities. Marketing synergies with existing small-business channels.

Breakout

Apply the Framework

Apex Robotics is considering a **$45 million** investment to deploy autonomous robotic picking-and-packing systems across three warehouses. The VP of Operations projects a 22% IRR based on $8–10M in annual labor savings plus $2M in error-reduction savings.

Complications: Apex's products require high-SKU-variety, small-batch picking — conditions where autonomous systems have historically underperformed. The robot vendor is a startup with only two prior installations at this scale. The VP's bonus is tied to cost-reduction targets.

In your group, work through all five issues (15 minutes)

1. Does the story make sense? What competitive advantage is Apex claiming?
2. What are the key risks? Which could invalidate the investment thesis entirely?
3. What financing considerations are relevant given the $45M would use 60% of Apex's credit facility?
4. Is the investment likely to be dilutive or accretive in the near term?
5. Can this be staged? What follow-on options exist if it succeeds — or fails?

The Investment Evaluation Process

Turning complexity into disciplined decision-making

Three Phases, Eight Steps

Phase	Steps	Purpose
Phase I: Origination & Analysis	1. Strategic assessment	Does the story hold up?
	2. Estimate investment value (crunch the numbers)	Build the financial case
	3. Prepare report and recommendation	Document assumptions
Phase II: Managerial Review	4. Evaluate strategic assumptions	Independent check on the story
	5. Review NPV methods and assumptions	Scrutinize the numbers
	6. Adjust for bias; formulate recommendation	Calibrate for overconfidence
Phase III: Decision & Approval	7. Make a decision	Accept, reject, or defer
	8. Seek final managerial / board approval	Commit resources

Step 1: Strategic Assessment

Every investment has an underlying “value proposition.”

- Initiated by the business development group
- Screens for competitive advantage
- Stops bad ideas early — before expensive analysis

Step 2: Crunch the Numbers

Estimate the investment’s value using:

- Discounted cash flow (DCF) analysis
- Market-based multiples
- Real options where relevant
- The financial model must be consistent with the strategic story

Step 3: Prepare the Report

The investment analysis report contains:

- Strategic rationale and competitive advantage assessment
- NPV estimate and key assumptions
- Supporting cash flow and cost projections
- Risk factors and

Phase II: Managerial Review

The review committee acts as a **skeptical boss** — its job is to ferret out bias.

What the Committee Does

- Step 4: Independently evaluates the strategic story — does it hold up to scrutiny?
- Step 5: Reviews financial methodology and assumptions — are the projections realistic?
- Step 6: Adjusts for optimism bias — projects are routinely 50% over budget and rarely completed more than 10% under budget

Why It Exists

Project champions have incentives to get deals approved. Even well-intentioned analysts are subject to overconfidence bias.

The committee is staffed by people who **did not** originate the project — independence is the point.

It must be funded and empowered independently. A committee that lacks authority to say no adds no value.

Incentive Bias

- Bonuses tied to “deals done” or cost targets
- Career advancement linked to project approval
- Champions who built the model personally invested in its conclusions
- Even honest people shade estimates when their job depends on the outcome

Psychological Bias

- Overconfidence: most people believe they are above-average drivers
- Self-attribution bias: success is skill, failure is bad luck
- The most optimistic, hardest-working managers get promoted — and become the most overconfident executives
- Good intentions do not overcome systematic optimism

Building Valuation Models with AI

Claude as your financial modeling assistant

Step 2 of Phase I is “crunch the numbers.” In practice, that means building financial models in Excel — pro forma income statements, FCF schedules, NPV calculations, sensitivity tables.

The Traditional Approach

- Analyst spends hours building the model from scratch
- Formatting, formulas, and layout are all manual
- Sensitivity analysis requires building additional tables by hand
- Errors are common and hard to detect

With Claude

- Describe the model in plain English — Claude builds it
- Professional formatting, labeled inputs, linked formulas
- Sensitivity tables generated automatically
- You focus on the **assumptions and the story**, not the spreadsheet mechanics

Claude + Excel: Two Approaches

Claude Generates the File

Ask Claude (Chat, Code, or Cowork mode) to create an `.xlsx` file from scratch.

Claude uses Python to build the workbook — fonts, colors, formulas, charts, and multiple sheets — and delivers a file you open directly in Excel.

Best for: building a new model from a description or template.

Claude Add-In for Excel

Install the Claude add-in from the Microsoft Office Add-ins store. A chat panel appears directly inside Excel.

Describe what you want — formulas, tables, charts, what-if analysis — and Claude builds it in the active sheet.

Best for: iterating on an existing model, adding sensitivity tables, fixing formula errors.

Both approaches produce the same result. The add-in is more interactive; the file-generation approach is better

Example: Building a DCF Model

What you tell Claude

```
I need an Excel workbook to evaluate a capital expenditure proposal. The investment is $547,000
in a materials-handling system with a 5-year life, straight-line depreciation to zero salvage
value, and a 40% tax rate. Annual pre-tax benefits are $343,800. Create: (1) an Inputs sheet
with clearly labeled yellow input cells, (2) a Cash Flows sheet showing NOPAT, depreciation
add-back, and free cash flow for each year, (3) a Results sheet with NPV, IRR, and payback
period using a 12% discount rate, and (4) a Sensitivity table showing NPV for discount rates
from 8% to 20% in 2% steps and annual benefits from $240,000 to $450,000 in $30,000 steps. Use
professional formatting with a dark navy header row.
```

One paragraph. No formulas, no column letters, no cell references — Claude figures out the structure. The output is a complete, working `.xlsx` file.

The Claude add-in puts a chat panel directly inside Excel. You describe what you want and Claude builds it in the active sheet — no Python, no terminal, no file downloads.

Installation

1. Open Excel
2. Go to Insert → Add-ins → Get Add-ins
3. Search for “Claude”
4. Install the Anthropic Claude add-in
5. Sign in with your Anthropic account

Good Uses in Valuation Work

- “Add a sensitivity table to this sheet for discount rates 8–16%”
- “Fix the IRR formula in cell B12 — it’s returning an error”
- “Add a chart showing NPV vs. discount rate”
- “Format this table with alternating

Still Your Job

- Assess whether the strategic story makes sense
- Judge whether the assumptions are realistic
- Identify the risks that matter most
- Decide whether to recommend the investment

Claude builds the model. **You make the judgment calls.**

Watch Out For

- Plausible-looking formulas that are wrong
- Sensitivity ranges that don’t match your actual uncertainty
- Models that are technically correct but built on assumptions Claude invented

Always **verify the outputs** — check that NPV matches what you’d calculate by hand for a simple case before trusting the full model.

Breakout

Build a Valuation Model with Claude

CP3 Pharmaceuticals: Build the Model

CP3 Pharmaceuticals is evaluating a **$547,000** materials-handling system. Annual pre-tax benefits: $300,000 in scrap revenue + $35,000 labor savings + $8,800 recycling savings = **$343,800**. Depreciation: straight-line over 5 years, zero salvage. Tax rate: 40%.

The strategic planning committee wants to see a sensitivity analysis before approving.

In your group (15 minutes)

1. Use Claude (chat or add-in) to build an Excel model with: an Inputs sheet, a Cash Flows sheet, a Results sheet (NPV, IRR, payback), and a sensitivity table (discount rate 8–20% × annual benefits ±30%)
2. At what discount rate does the project turn NPV-negative?
3. If benefits come in 20% below projection, is the project still attractive at a 12% discount rate?
4. The committee suspects optimism bias. What adjustment to the discount rate or cash flows would you make as a skeptical reviewer, and does the recommendation change?

Summary

Key Takeaways

Valuation is the Foundation

- Both project and enterprise decisions require the same core framework
- Value = present value of expected cash flows minus cost
- Over half of large investments fail — disciplined process improves the odds

Five Issues Every Time

- Story, risks, financing, earnings, flexibility
- The qualitative assessment must be consistent with the financial model
- A compelling story with a negative NPV is still a bad investment

Process Combats Bias

- Three phases, eight steps
- Phase II independence is critical — the committee must be empowered to say no
- AI tools speed up Step 2 (crunch the numbers) but do not replace judgment

Coming Up: Chapter 2

Forecasting and Valuing Cash Flows

- Defining free cash flow (FCF)
- Incremental cash flows — what counts and what doesn't
- Sunk costs, cannibalization, working capital
- Building pro forma cash flow statements

NPV and IRR

- Discounting FCF to estimate project value
- Net present value and internal rate of return
- Mutually exclusive projects
- Mini-cases: Meridian Pharma, Ridgeway Furniture, NovaBrew